

HeatShift 2035: Which Homes and Places Should Switch to Heat Pumps First?

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Code Available At: github.com/NiharJani2002/HeatShift-2035

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1 The problem I wanted to solve

Councils and community housing providers have a simple question and no cheap way to answer it: if we can only upgrade some homes this year, which ones should come first? Swapping a gas heater for a heat pump saves money and carbon, but how much depends on the climate, the shape of the house, the price of gas and electricity, and how well the heat pump performs on cold days. At the same time, some places are much more exposed to heatwaves, and a home that is hard to keep cool is a health risk, not just a bill.

I built HeatShift 2035 to bring those two things together. The main priority is electrification (the COP31 goal of reaching 35% electrified by 2035). The second is resilient cities and buildings. The third is awareness: every result ends in a plain-language action card that a household can read in under a minute.

I want to be upfront about one thing. I did not build a black box that outputs a rank and asks for trust. Every price, efficiency and emissions factor is a slider in the sidebar, every saving comes with a range rather than one number, and every statistical method in the project is tested on simulated data where I know the right answer. That last part is the thing I am proudest of, and it is why this report spends most of its time on testing.

2 Data

The core of the tool uses two Kaggle datasets. The first is the Rain in Australia weather data (`weatherAUS.csv`), which gives daily minimum and maximum temperature for 49 stations, roughly 2007 to 2017. The second is the UCI Energy Efficiency dataset (`ENB2012_data.csv`) from Tsanas and Xifara, which has 768 simulated building designs. Each design has eight inputs (compactness, surface area, wall area, roof area, height, orientation, glazing area and glazing distribution) and two outputs, a heating load and a cooling load.

The other tabs use further Kaggle datasets: EV population, global methane, flood prediction, crop recommendation, global sustainable energy, OWID world energy, the World Bank “What a Waste” data, Berkeley Earth temperatures, and crop yields. The code finds each file by its column names rather than its file name, so it works with whatever the user has downloaded and shows a polite message if a file is missing.

3 The core model

3.1 Climate side

For each station I fill short gaps in the temperature record, take the daily mean T , and compute heating degree-days with an 18°C base:

$$\text{HDD} = \sum_t \max(0, 18 - T_t). \quad (1)$$

A heat pump is not equally efficient all year. Its coefficient of performance (COP) falls when it is colder outside. I model it as a straight line through the rated COP at 7°C, with a slope of 0.10 per degree, clipped to the range [1.8, 5]. The seasonal COP is then a degree-day weighted average that gives the cold days the weight they deserve:

$$\text{SCOP} = \frac{\sum_t \text{HDD}_t}{\sum_t \text{HDD}_t / \text{COP}(T_t)}, \quad \text{COP}(T) = \text{clip}(\text{COP}_7 + \delta + 0.10 (T - 7), 1.8, 5). \quad (2)$$

The term δ is a COP uncertainty shift that I vary between -0.8 and $+0.8$.

For heatwaves I use the Excess Heat Factor of Nairn and Fawcett. A three-day mean temperature is compared with the last 30 days (acclimatisation) and with the 95th percentile of the station (significance). A day counts as a heatwave day when the factor is positive, and as a severe day when it is in the top 15% of positive values. I have to be honest that my local record is about ten years, not the thirty years the Bureau uses, so I treat heatwave days as a relative exposure measure between places, not an official count.

3.2 Building side

The building loads come from a gradient boosting model (300 trees, depth 3, learning rate 0.05, subsample 0.8). The ENB2012 loads are simulated for one reference climate, so I treat them as annual loads in a chosen anchor climate (Melbourne by default) and scale them to other places by degree-days:

$$D = L_{\text{heat}} \cdot A \cdot \frac{\text{HDD}}{\text{HDD}_{\text{ref}}}, \quad (3)$$

where A is the heated floor area. Absolute dollar values depend on this choice, but the ranking of places does not, because the scaling is the same for every place.

3.3 Money, carbon and triage

With gas heater efficiency η , gas price p_g per kWh of gas and electricity price p_e , the yearly saving and avoided emissions are

$$\text{Saving} = \frac{D}{\eta} p_g - \frac{D}{\text{SCOP}} p_e, \quad \Delta\text{CO}_2 = \frac{D}{\eta} e_{\text{gas}} - \frac{D}{\text{SCOP}} e_{\text{grid}}. \quad (4)$$

I report ΔCO_2 twice, once with today's grid factor and once with a cleaner 2035 grid, because a heat pump gets better for the climate every year the grid gets cleaner.

To decide who goes first I combine two percentile ranks, savings and heatwave days, using a weight w that the user sets:

$$\text{Triage} = 100 (w \cdot \text{rank}(\text{Saving}) + (1 - w) \cdot \text{rank}(\text{Heatwave days})). \quad (5)$$

Each place also gets a recommended first move: a reverse-cycle heat pump if it is high on both, electrify heating first if savings dominate, and heat-resilience first (cooling access, shading) if heat risk dominates.

3.4 Uncertainty

Every saving shown to the user is the middle of 3,000 random draws, with a P10 to P90 range and the probability that the saving is above zero. The draws vary electricity price (lognormal, spread 0.2), gas price (0.25), heater efficiency (uniform 0.7 to 0.9), a load multiplier (0.3) and the COP shift. I would rather show “likely \$300 to \$900” than a single confident number I cannot defend.

4 Does it work? Two honest tests

Unseen building shapes. The ENB2012 data has many near-identical designs. If I shuffle and split randomly, the model sees almost the same house in training and in testing, and the score looks great for the wrong reason. So the app reports two cross-validation results side by side: random five-fold, and a grouped test that holds out whole building shapes (grouped by compactness). The grouped result is the one I trust, and the app compares both with a baseline that always guesses the average. These numbers are computed live when the app starts, so I do not hard-code them into this report.

Stability over time. I split the weather record into 2007 to 2012 and 2013 to 2017, rank the places in each half, and report the Spearman rank agreement for heating payoff and for heatwave days, plus how many of the top ten places stay in the top ten. If the triage changed completely between halves, it would not be worth acting on.

5 The ten mathematical methods

Beyond the core model I wrote ten methods in `mathcore.py`, using only NumPy and SciPy. For each one I wrote a test in `proofs.py` that simulates a world where the true answer is known and checks that the method finds it. None of these tests use Kaggle data. Here is what each method does and what the test showed (numbers from running the script).

1. Conformal prediction intervals. A normal prediction interval built from training residuals is too narrow for a new building shape. I get out-of-fold residuals by leaving whole groups out, then take the $\lceil (n+1)(1-\alpha) \rceil / n$ quantile of the absolute residuals as the interval half-width. *Test:* for a 90% target, coverage on unseen shapes was 88.7%, against 71.4% for the naive interval.

2. Empirical-Bayes shrinkage. Heatwave counts for one place are noisy. I model counts as $y_i \sim \text{Poisson}(\lambda_i e_i)$ with $\lambda_i \sim \text{Gamma}(a, b)$, estimate (a, b) from the marginal negative-binomial likelihood, and use the posterior mean $(a + y_i)/(b + e_i)$. This also gives probabilities for ranks. *Test:* for rare events (about one a year), shrinkage beat raw counts in 95% of 300 simulated worlds and cut squared error by 31%, with 87.9% interval coverage. For common events the gain was only 8%, which is correct, since there is little noise to borrow from.

3. SMAA (stochastic multicriteria acceptability analysis). Instead of picking one triage weight, I draw weights uniformly from all possibilities and count how often each place lands at each rank. *Test:* against an exact integral for two criteria, the error fell from 0.0143 to 0.0012 as draws increased, with a log-log slope of -0.44 (theory says -0.5).

4. CVaR linear programme. A council has a budget and wants the most emissions saved, without the worst 10% of scenarios losing money. I use the Rockafellar–Uryasev linearisation: maximise abatement subject to the budget, a cap per option, and a tail constraint

$$\zeta + \frac{1}{(1-\alpha)N} \sum_k u_k \leq 0, \quad u_k \geq -S_k x - \zeta, \quad u_k \geq 0. \quad (6)$$

Test: the optimum (114,646) beat the best of 42,000 random feasible plans (112,515); the tail constraint was met; and when the tail was slack it matched the greedy knapsack answer exactly (114,790 against 114,790).

5. Extreme values and trend. I fit a generalised Pareto distribution to heat exceedances by probability-weighted moments and compute return levels

$$x_T = u + \frac{\sigma}{\xi} ((\lambda T)^\xi - 1), \quad (7)$$

with a bootstrap interval. For trends I use Mann–Kendall with the variance inflated by $(1 + \rho)/(1 - \rho)$ for lag-1 autocorrelation, and Theil–Sen slopes. *Test:* the 50-year return level interval covered the truth 87.7% of the time (90% target). Under autocorrelated noise with no trend, the naive trend test raised false alarms 27.8% of the time; the corrected one 7.0%, while still detecting a real trend 99% of the time. Theil–Sen had a median error of 0.0039 against 0.0183 for OLS when 12% of points were outliers.

6. Fixed-effects panel regression. To measure how warming affects crop yield, I demean within each place and use cluster-robust standard errors. Hot places tend to have low baseline yields, which badly biases a pooled regression. *Test:* with a true effect of 0.50, pooled OLS was off by -0.99 and the fixed-effects estimate by -0.001 . Ninety-five percent interval coverage was 92.7% with clustering against 84.3% without.

7. Sobol sensitivity, landfill decay and Dirichlet fit. Sobol indices tell me which input uncertainty matters most for landfill methane. The landfill model is a first-order decay model that conserves mass: what is buried is either decayed or still in the pile, so $\text{acc}_t + \text{dec}_t = \text{acc}_{t-1} + \text{dd}_t$ exactly. *Test:* Sobol indices matched the analytic Ishigami benchmark within 0.012; the mass balance error was 1.8×10^{-15} ; and the Dirichlet fit for waste composition had a worst relative error of 5.8%.

8. Markov chain and Gamma lifetimes. For circular industry I model a material moving through in-use, collected, sorted and reprocessed states. The expected number of uses per tonne comes from the fundamental matrix $N = (I - Q)^{-1}$. With collection 0.8, sorting 0.7 and yield 0.9 the circular rate is 0.504, so a tonne is used $1/(1 - 0.504) \approx 2.016$ times. With Gamma-distributed product lifetimes and growing demand there is also a closed form, $\rho(1 + \theta \ln(1 + g))^{-k}$. *Test:* theory gave 2.016 and a 200,000-parcel simulation gave 2.014; the closed form was within 0.5% of an item-level stock-flow simulation.

9. Logistic S-curve forecast. Adoption curves are S-shaped, so I regress $\text{logit}(s/100)$ on year. The prediction is a t-distribution, which gives an exact probability of a country reaching its target. *Test:* on 3,000 simulated countries the probability integral transform was uniform (KS $p = 0.19$), 90% intervals covered 89.7%, and the Brier score was 0.146 against 0.140 for perfect calibration.

10. Thompson sampling. For awareness messages, I want to learn which message works without wasting households on bad ones. *Test:* after 3,000 households, cumulative regret was 45 for Thompson sampling, 63 for epsilon-greedy and 247 for an equal-split A/B test. The best message got 84% of households. With ten times more households, Thompson regret grew only 3.1 times, while the A/B regret grew 10 times.

6 The other priorities

The app has a tab for each of the other COP31 priorities. The electrification tab adds a whole-home passport (heat pump plus EV against petrol) and uses the Washington State EV registry as an adoption-speed signal, not as an Australian forecast. The waste tab runs the landfill model to 2035 with food diversion and gas capture. The industry tab estimates circular material use. The flood tab ranks which levers a council should pull first. The farming tab asks whether today’s best crop still suits a place in a hotter, drier climate, and includes the fixed-effects estimate of the cost of warming for each crop. There is also a short teach-back quiz, a COP31 scorecard with live numbers, a data library that discloses every dataset in use, and a judges’ guide that maps each judging criterion to evidence in the app.

Method	Quantity checked	Result
Conformal	coverage, 90% target	88.7% (naive 71.4%)
Empirical Bayes	MSE reduction, rare events	31%
SMAA	error slope vs. draws	-0.44 (theory -0.5)
CVaR LP	optimum vs. greedy knapsack	114,790 vs. 114,790
Extreme values	false alarms, trend test	27.8% → 7.0%
Fixed effects	bias in true effect 0.50	-0.001 (OLS -0.99)
Sobol / landfill	mass balance error	1.8×10^{-15}
Markov / Gamma	expected uses, theory vs. sim.	2.016 vs. 2.014
S-curve	90% interval coverage	89.7%
Thompson	regret, 3,000 households	45 (A/B 247)

Table 1: Summary of the ten proofs. All ten pass when I run `python proofs.py`.

7 Limitations

- The ENB2012 loads are simulated for a single reference climate, not metered from real homes. Scaling by degree-days is a simplification.
- The weather record covers about 2007 to 2017, so it misses recent extremes, and my heatwave days use a ten-year local record rather than the official thirty.
- Some places have no gas network, so the gas-to-heat-pump comparison does not apply there.
- Prices, the COP curve and grid factors are assumptions. They are editable, but they are still assumptions. The gas emissions factor in the code is marked to be checked against the national inventory factors.
- Station coordinates on the map are approximate.
- Passing a proof means the method works when its assumptions hold. It does not prove the real world follows them. For example, conformal coverage needs building shapes to be exchangeable.

8 From prototype to pilot

The real barrier is that councils have no cheap way to decide where electrification pays off fastest. A realistic next step is an 8 to 12 week pilot with about 30 homes in two contrasting climates. I would log current smart-meter data, run HeatShift’s predictions before any installation, install heat pumps in half the homes first in a stepped rollout, and then compare predicted and metered heating energy. My targets are a median prediction error below 25%, a triage rank that agrees with installer-reported priority, and 80% of households able to state their next step after reading the action card. I would also record bills saved and kilograms of CO₂e avoided.

9 Conclusion

HeatShift 2035 turns two open datasets into a practical answer for a council: where to start, which home, and what to tell the household. It is not perfect, and I have tried to say clearly where it is weak. What I can say with confidence is that the maths underneath is tested against known truth, and that anyone can change my assumptions and watch the answer move.

How to run it. Install `streamlit pandas numpy scikit-learn scipy plotly kagglehub`, put the Kaggle CSVs in the project folder or `data`, then run `streamlit run app.py`. To check the maths, run `python proofs.py`.